

Quarter I review paper · Algebra

Grade 11

40 minutes 100 marks 13 questions + bonus No calculator

3–4 ext	Limits at infinity — extension	Algebra 11, §1.2 · P1 · 7.1 · new on this paper
5–6	The derivative of a function	Algebra 11, §1.3 · P1 · 7.1–7.2
7–9	The rules of differentiation	Algebra 11, §1.4 · P1 · 7.2–7.4
10–12	The derivative of a composite function	Algebra 11, §1.5 · P1 · 7.5
15–16	The modulus function	P1 · 1.6 · P2 · 1.1–1.3
17–18	The equations of the tangent and the normal	Algebra 11, §1.6 · P1 · 7.6
19–22	Investigating a function with the derivative	Algebra 11, §1.7 · P1 · 7.7–7.8

- Answer on separate paper, and show your working — method carries marks.
- State the nature of every stationary point, and the test you used.

What the paper weighs

TOPIC	MARKS	SHARE
Lessons 3–4, extension · Limits at infinity	13	13%
Lessons 5–6 · The derivative of a function	8	8%
Lessons 7–9 · The rules of differentiation	13	13%
Lessons 10–12 · The derivative of a composite function	13	13%
Lessons 15–16 · The modulus function	10	10%
Lessons 17–18 · The equations of the tangent and the normal	23	23%
Lessons 19–22 · Investigating a function with the derivative	20	20%

Nothing on this paper needs lessons 1–2, 3–4 or 23–25. Limits at infinity are new, and carry 13 marks. The tangent and the normal carry the largest share because question 13 asks for both from a composite function, so it also examines the chain rule.

A Section A · Five marks each

EASY 30 marks

Short answers — one or two lines each.

1 Differentiate: $y = 4x^3 - 5x + 7$ [5]

ANSWER

$$\frac{dy}{dx} = 12x^2 - 5$$

WORKING

Power rule term by term; the derivative of the constant 7 is zero.

2 Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$ [5]

ANSWER

3

WORKING

Divide top and bottom by x^2 : $\frac{3 + 2/x}{1 - 5/x^2}$. Both $\frac{2}{x}$ and $\frac{5}{x^2}$ tend to 0, leaving $\frac{3}{1} = 3$.

Equal degrees, so the answer is the ratio of the leading coefficients — but the division has to be shown, not quoted.

3 Differentiate: $y = (3x + 1)^5$ [5]

ANSWER

$$\frac{dy}{dx} = 15(3x + 1)^4$$

WORKING

Chain rule: $5(3x + 1)^4$ times the derivative of the inside, which is 3.

4 Solve: $|2x - 6| = 10$ [5]

ANSWER

$$x = 8 \text{ or } x = -2$$

WORKING

$2x - 6 = 10$ gives $x = 8$; $2x - 6 = -10$ gives $x = -2$.

One answer only means the negative branch was forgotten.

5 Find the stationary point of $y = x^2 - 6x + 5$ and state its nature. [5]

ANSWER

$(3, -4)$, a minimum

WORKING

$\frac{dy}{dx} = 2x - 6 = 0$ gives $x = 3$, and $y = 9 - 18 + 5 = -4$. $\frac{d^2y}{dx^2} = 2 > 0$, so it is a minimum.

6 Solve: $|x - 3| < 5$ [5]

ANSWER

$$-2 < x < 8$$

WORKING

The distance from x to 3 is less than 5, so $-5 < x - 3 < 5$. Adding 3 throughout gives $-2 < x < 8$.

B Section B · Eight marks each

Method marks are available even when the final answer is wrong.

MEDIUM

40 marks

7 Evaluate, showing the division in each case:

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 1}{5x^2 + 4}$ [3]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 - 2}$ [2]

(c) Explain why $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{4x^2 - 1}$ does not exist. [3]

ANSWER

(a) $\frac{2}{5}$ (b) 0 (c) it increases without bound

WORKING

(a) Divide by x^2 : $\frac{2 - 3/x + 1/x^2}{5 + 4/x^2} \rightarrow \frac{2}{5}$.

(b) Divide by x^3 : $\frac{1/x + 1/x^3}{1 - 2/x^3} \rightarrow \frac{0}{1} = 0$.

(c) Divide by x^2 : $\frac{x + 2/x}{4 - 1/x^2}$. The denominator tends to 4, but the numerator contains x , which grows without bound — so the quotient grows without bound and no finite limit exists.

The three parts are the three degree cases: top = bottom, top < bottom, top > bottom. A student who has only learnt “compare the degrees” can state the answers but cannot earn the method marks.

8 $f(x) = x^2 - 3x$.

(a) Show that $\frac{f(2+h) - f(2)}{h} = h + 1$. [5]

(b) Hence write down $f'(2)$. [3]

ANSWER

(a) shown below (b) $f'(2) = 1$

WORKING

(a) $f(2+h) = (2+h)^2 - 3(2+h) = 4 + 4h + h^2 - 6 - 3h = h^2 + h - 2$, and $f(2) = 4 - 6 = -2$. So the numerator is $h^2 + h$, and dividing by h gives $h + 1$ — legal because $h \neq 0$ throughout.

(b) As $h \rightarrow 0$, $h + 1 \rightarrow 1$, so $f'(2) = 1$.

Cancelling the h before noting $h \neq 0$ is the slip to look for. Checking with the rules, $f'(x) = 2x - 3$ and $f'(2) = 1$, is worth a mark but is not the method asked for.

- 9 Differentiate $y = \frac{2x-1}{x^2+3}$ and find the gradient of the curve at $x = 1$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}; \text{ gradient} = \frac{3}{8}$$

WORKING

Quotient rule with $u = 2x - 1$, $v = x^2 + 3$, so $u' = 2$ and $v' = 2x$:

$$\frac{dy}{dx} = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2} = \frac{2x^2 + 6 - 4x^2 + 2x}{(x^2 + 3)^2} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}.$$

$$\text{At } x = 1: \frac{-2 + 2 + 6}{4^2} = \frac{6}{16} = \frac{3}{8}.$$

The expansion $-(2x - 1)(2x) = -4x^2 + 2x$ is where the sign is usually lost.

- 10 The curve $y = x^3 - 4x$. Find the equation of the tangent and the equation of the normal at the point where $x = 2$. [8]

ANSWER

$$\text{tangent } y = 8x - 16; \text{ normal } x + 8y = 2$$

WORKING

At $x = 2$, $y = 8 - 8 = 0$, so the point is $(2, 0)$. $\frac{dy}{dx} = 3x^2 - 4$, which is 8 at $x = 2$.

Tangent: $y - 0 = 8(x - 2)$, so $y = 8x - 16$.

Normal gradient $= -\frac{1}{8}$: $y - 0 = -\frac{1}{8}(x - 2)$, so $8y = -x + 2$, that is $x + 8y = 2$.

Using -8 rather than $-\frac{1}{8}$ for the normal is the standard error; the product of the two gradients must be -1 .

- 11 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 9}}; \text{ at } x = 4 \text{ it is } \frac{4}{5}$$

WORKING

Write $y = (x^2 + 9)^{1/2}$. Chain rule: $\frac{dy}{dx} = \frac{1}{2}(x^2 + 9)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 9}}$.

At $x = 4$: $\sqrt{16 + 9} = 5$, so the value is $\frac{4}{5}$.

The factor $2x$ from the inside function is the mark most often dropped.

C Section C • Fifteen marks each

Full method required.

HARD

30 marks

12 $y = x^3 - 6x^2 + 9x$.

- (a) Find the stationary points and determine the nature of each. [6]
(b) State the intervals on which the function is increasing and decreasing, and sketch the curve. [5]
(c) Find the values of k for which $x^3 - 6x^2 + 9x = k$ has three distinct solutions. [4]

ANSWER

- (a) (1, 4) maximum, (3, 0) minimum (b) increasing for $x < 1$ and $x > 3$, decreasing for $1 < x < 3$
(c) $0 < k < 4$

WORKING

(a) $\frac{dy}{dx} = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, zero at $x = 1$ and $x = 3$. Then $y(1) = 1 - 6 + 9 = 4$ and $y(3) = 27 - 54 + 27 = 0$.

Second derivative $\frac{d^2y}{dx^2} = 6x - 12$: at $x = 1$ it is $-6 < 0$, a maximum; at $x = 3$ it is $6 > 0$, a minimum.

(b) $3(x-1)(x-3) > 0$ for $x < 1$ and for $x > 3$, so the function increases there, and < 0 for $1 < x < 3$, so it decreases between the two stationary points. The curve passes through the origin, since $y = x(x-3)^2$, and rises without bound on the right.

(c) The solutions of $y = k$ are where the horizontal line $y = k$ cuts the curve. It cuts three times exactly when k lies strictly between the minimum value 0 and the maximum value 4, so $0 < k < 4$.

A sign chart earns the nature marks just as well as the second derivative. Part (c) is the sketch being used, not a new technique — a student who drew the curve in (b) can read the answer off it. The inequalities must be strict: at $k = 0$ and $k = 4$ the line passes through a stationary point and there are only two distinct solutions.

13 The curve $y = \sqrt{2x+7}$.

- (a) Find $\frac{dy}{dx}$. [4]
(b) Find the equation of the tangent at the point where $x = 1$. [6]
(c) Find the equation of the normal at that point, and the coordinates of the point where it crosses the x -axis. [5]

ANSWER

- (a) $\frac{dy}{dx} = \frac{1}{\sqrt{2x+7}}$ (b) $3y = x + 8$ (c) $y = 6 - 3x$, crossing at (2, 0)

WORKING

(a) $y = (2x+7)^{1/2}$, so $\frac{dy}{dx} = \frac{1}{2}(2x+7)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x+7}}$.

(b) At $x = 1$: $y = \sqrt{9} = 3$ and $\frac{dy}{dx} = \frac{1}{3}$. So $y - 3 = \frac{1}{3}(x - 1)$, which tidies to $3y = x + 8$.

(c) The normal gradient is -3 , so $y - 3 = -3(x - 1)$, that is $y = 6 - 3x$. It meets the x -axis where $y = 0$, so $x = 2$ and the point is (2, 0).

The $\cdot 2$ from the inside function in (a) is the mark most often dropped, and it carries through both later parts.

Leaving the tangent as $y = \frac{1}{3}x + \frac{8}{3}$ is not penalised.

Bonus



BONUS

+10 marks

Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

- 14 Find the equations of the two tangents to $y = x^2$ that pass through the point $(0, -4)$. [10]

ANSWER

$$y = 4x - 4 \text{ and } y = -4x - 4$$

WORKING

Let the point of contact be (t, t^2) . Since $\frac{dy}{dx} = 2x$, the tangent there has gradient $2t$, so its equation is

$$y - t^2 = 2t(x - t), \text{ that is } y = 2tx - t^2.$$

It passes through $(0, -4)$, so $-4 = -t^2$, giving $t = 2$ or $t = -2$.

The two tangents are $y = 4x - 4$ and $y = -4x - 4$.

The point $(0, -4)$ is *not* on the curve, so there is no single point to differentiate at — the contact point has to be carried as an unknown. That is the whole difficulty, and it is why this is the bonus.

END OF MARK SCHEME